

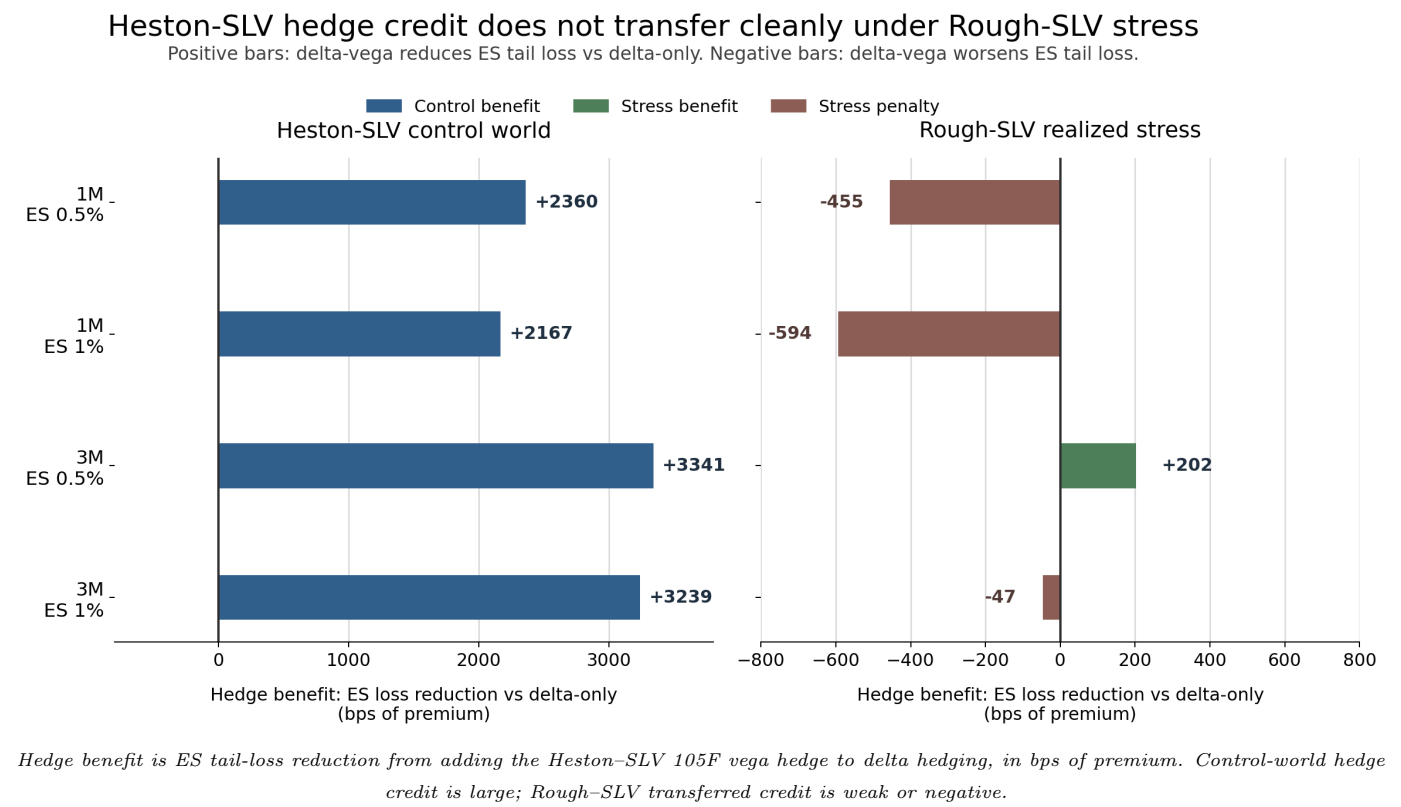
When Smile Fit Overstates Hedge Credit

FX Hedge-Transfer and XVA Diagnostics under Rough Realized Dynamics

Core claim. A calibrated FX volatility model can fit the same vanilla smile and still overstate the hedge credit that should be assigned under stressed realized dynamics. This project tests that effect by holding the static vanilla surface fixed, using Heston-SLV as the pricing and hedge model, and stressing realized dynamics with Rough-SLV.

Controlled design. The experiment separates smile fit from hedge reliability.

same static smile + same Heston-SLV hedge + different realized dynamics ⇒ hedge-transfer diagnostic



Main hedge-transfer result. The Heston-SLV delta-vega hedge earns large ES tail-loss credit in its own Heston-SLV control world, but little or none of that credit transfers under Rough-SLV realized dynamics.

Maturity	Metric	Heston control	Rough stress	Desk read
1M	ES0.5	+52.9%	-30.9%	Negative transfer; hedge worsens stress tail loss
1M	ES1.0	+51.8%	-46.8%	Severe transfer failure
3M	ES0.5	+41.0%	+3.0%	Almost no useful transfer
3M	ES1.0	+42.6%	-3.0%	Deep-tail transfer failure

Desk metric. The result is translated into a stress hedge-credit haircut:

Transfer Ratio = $\frac{\text{Rough-SLV hedge benefit}}{\text{Heston-SLV control hedge benefit}}$, Haircut = $1 - \max(\text{Transfer Ratio}, 0)$.

Desk implication. A hedge that earns 40–50% ES reduction in its own model world should not automatically receive 40–50% stress hedge credit. If transferred hedge benefit is near zero or negative, stress hedge credit should be heavily haircut.

XVA Materiality, Scope, and Limitations

Exposure / XVA extension. The hedge-transfer block is the main result. The XVA block tests whether the same volatility-dynamics misspecification propagates into future exposure timing and CVA-style materiality. It is a materiality diagnostic, not a production CVA engine.

Maturity	XVA read	Desk action
1M	Low materiality	No escalation from XVA alone
3M	Positive monitor-worthy signal	Monitor and attribute
1Y	Strongest exposure-timing signal	Review / reserve discussion if portfolio materiality justifies it

Controlled limitations. The surface is synthetic and used for model-risk isolation. Rough-SLV is used as adverse realized-dynamics stress, not claimed as the true market model. Clean hedge-transfer evidence is concentrated at 1M and 3M; 1Y hedge evidence is treated as limitation, while 1Y is used mainly for exposure / XVA materiality.

Public scope. This repository contains selected result summaries, figures, methodology notes, source-code excerpts, and a reduced reproducibility demo. The full research implementation, production-scale Monte Carlo pipeline, frozen hedge surfaces, seed packs, pathwise hedge P&L arrays, and validation logs are not released publicly.

Bottom line. Same smile. Same Heston-SLV hedge. Different realized dynamics. Different hedge credit.